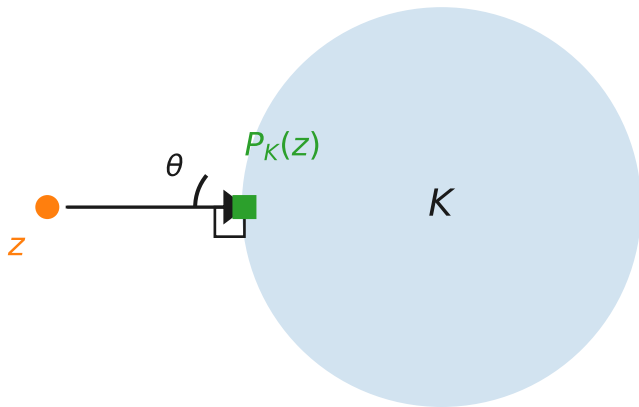




$P_K(z)$ is the closest point in K to z

$$(z - P_K(z))^T(x - P_K(z)) \geq 0 \text{ for all } x \in K$$